

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	i		K ⁺ (mainly) transported (upwards) in xylem (1) K ⁺ only moves {laterally / sideways} into phloem from xylem/ K ⁺ does not move up or down in the phloem (1)			2	2		
		ii		To show that the waxed paper stops the movement/ owtte (1) Shows that separating the xylem and phloem (and then putting back into contact) does not affect movement (1) <i>If when separated they return back together</i> Values would be the same at all points (1) <i>If stay separated</i> Expect values to be the same in xylem and phloem as in previous experiment (1)			2	2		2
	(b)	i		No / little transport in xylem (1) transport in phloem in both directions (1)			2	2		
		ii		Analyse samples from (above and) below the lower leaf (1) If both positive then movement in both directions (1)			2	2		

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	(c)			Any 5 from: 1. In summer, leaves {photosynthesise / make sugars} {loaded/ moved into} {phloem / sieve tubes} (1) 2. Water potential falls water absorbed by osmosis (1) Must be in phloem or sieve tubes 3. Reference to hydrostatic pressure (1) 4. Flow (in sieve tubes) from high to low pressure regions (1) 5. tubers (sucrose) converted into starch (1) 6. water potential rises + water lost (1) 7. (Sugars in tubers)allows (rapid) growth (in spring) / sugars used to form cellulose for growth/ sugars for respiration (1)	3	2		5		
				Question 6 total	3	2	8	13	0	2